

Principia XAU/USD Signal Platform

Checkpoint roadmap, tab plan, and under-\$200/month API budget

Planning intent

- Build a private, always-on Gold signal platform for discretionary trading first, with room to expand later.
- Keep hosting, training, jobs, and core operations on Google Cloud.
- Keep recurring software and API spend under \$200/month, with a preferred working range closer to \$100-125/month.
- Use the existing IBKR account wherever it reduces paid vendor count.
- Keep version 1 non-automated: the system informs the trade decision; it does not auto-execute.

Document type	Working roadmap and costed build plan
Starting asset	Gold (XAU/USD)
Primary cloud	Google Cloud Platform, region target europe-west2 (London)
Current foundation	Next.js frontend, FastAPI backend, Cloud Run, Cloud SQL, Secret Manager, Artifact Registry
Signal basis	Existing EMA/RSI checklist plus broader macro and microstructure framework
Target tabs	Home, Detailed Fundamentals, Detailed Indicators, Projection / Price Modeling / Forecasts, Backtesting
Date prepared	17 April 2026

This roadmap consolidates the uploaded build brief, technical summary, signal checklist, trading manual, and dashboard prototypes into one checkpoint-driven plan.

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1 What is being built

The platform is a private signal intelligence system for Gold. It should show current regime, current signal, confluence across indicators, alert history, and operational health in a format that is faster to read than a broker terminal. It should also grow into a proper research surface with fundamentals, indicators, forecasting, and backtesting in one place.

Core product position

Version 1 should remain an **internal discretionary tool**. The dashboard can inform entries, exits, and stand-aside decisions, but it should not auto-trade in the first release. That keeps technical complexity, operational risk, and compliance risk low while the process is still being validated.

Scope that is already clear

- Gold is the starting asset and the immediate priority.
- The platform needs live signal visibility, regime state, alerting, and a full history log.
- The dashboard must eventually expose deeper tabs for fundamentals, indicators, forecasts, and back-testing.
- Hosting, scheduled jobs, and model refresh workflows should all live on Google Cloud.
- Cost discipline matters from the beginning.

What should not be in version 1

- No auto-execution.
- No GPU-heavy model training.
- No BigQuery-first architecture.
- No separate premium data vendors unless they clearly unlock measurable value.
- No multi-asset expansion until Gold is stable and trusted.

2 Product structure and target tabs

Tab	Primary purpose	Must-have content	Main data dependencies
Home	Immediate trading read	Current regime, active BUY/SELL/HOLD state, confidence score, last alert, last regime switch, recent signals, system health, last model run, ingestion heartbeat	Signal engine, alerts table, health checks, latest bars

Tab	Primary purpose	Must-have content	Main data dependencies
Detailed Fundamentals	Macro context and event risk	Real yields, DXY, Fed stance, CPI/NFP/GDP/PCE board, central bank buying, COT, GLD flows, VIX, oil, GDX relative strength, geopolitical and rates news	FRED, BLS, BEA, CFTC, MarketAux, EODHD calendar, manual WGC/PBOC ingest
Detailed Indicators	Chart-state and intra-day execution layer	EMA alignment, RSI, ADX, VWAP, volume, session windows, London Fix markers, stop-hunt levels, spread and open-interest context, correlation check	IBKR bars, internal calculations, optional MT5 or TradingView redundancy
Projection / Price Modeling / Forecasts	Forward-looking probabilities and scenario output	Short-horizon forecast, expected move bands, scenario tree, feature importance, model agreement/disagreement, reason codes, narrative summary	Historical bars, macro features, event flags, news sentiment, model registry
Backtesting	Strategy validation and iteration	Strategy library, walk-forward runs, parameter sweeps, slippage assumptions, event filters, equity curves, drawdowns, trade distribution, downloadable logs	Historical price store, feature store, strategy definitions, cached news/event context

UI direction

The original brief used Streamlit for speed, but the technical execution summary shows the stack has already moved to **Next.js** + **FastAPI** + **Cloud Run** + **Cloud SQL**. The roadmap should keep that direction rather than revert, because the current foundation is more modular for tabs, auth, and later commercialization.

3 Recommended Google Cloud architecture

Target build shape

- **Frontend:** Next.js application on Cloud Run.
- **Backend API:** FastAPI service on Cloud Run.
- **Database:** PostgreSQL on Cloud SQL.
- **Scheduled work:** Cloud Run Jobs triggered by Cloud Scheduler.
- **Model artifacts and backtest exports:** Cloud Storage.
- **Images and containers:** Artifact Registry.
- **Secrets:** Secret Manager.
- **Broker and market integration:** IBKR API and historical bars where permissions allow.

- **Public macro data:** FRED, BLS, BEA, CFTC, plus manual WGC and PBOC ingest where needed.
- **Low-cost paid feeds:** one news feed with sentiment, one calendar feed, optional minimal IBKR market data add-on depending the chosen gold instrument.

Architecture chain

Data ingest connectors → signal engine and feature layer → Cloud SQL and Cloud Storage → FastAPI → Next.js UI → Telegram / optional email alerts

Cost control rules

- Use Cloud Run request-based services and let them scale to zero when idle.
- Use Cloud Run Jobs for training, summarization, and backtest refresh instead of an always-on VM.
- Keep all core services in the same region.
- Stay CPU-only for version 1 modeling.
- Cache historical bars and derived features so external requests do not repeat unnecessarily.
- Do not pay for a separate NLP vendor if MarketAux covers first-pass sentiment and Gemini only needs to summarize.
- Keep logging inside free allotments and avoid long custom retention.
- Keep Cloud SQL small until the dataset proves it needs more.

4 Checkpoint roadmap from the starting point

Checkpoint 0 - Scope lock and budget guardrails

Objective: freeze the version 1 shape so engineering, data, and cost decisions stop moving around.

Tasks

- Confirm the canonical traded instrument path: IBKR XAUUSD / IBKR USGOLD / COMEX futures via IBKR.
- Lock version 1 non-goals: no auto-trading, no GPU jobs, no multi-asset expansion, no mobile app.
- Freeze the five user-facing tabs and define what must exist in each tab before launch.
- Lock the recurring spend rule: hard ceiling under \$200/month, preferred operating range \$100-125/month.
- Decide the primary signal source: backend-computed signals from broker bars or MT5, with TradingView only as optional validation or redundancy.
- Confirm that the current Next.js/FastAPI/GCP foundation is the stack of record.

Exit state

- One written scope sheet.
- One accepted cost cap.
- One instrument map for Gold.
- One stack decision with no unresolved version 1 platform debate.

Checkpoint 1 - Google Cloud foundation

Objective: make the cloud base production-like before feature work spreads.

Tasks

- Create or confirm the production project and billing alerts.
- Standardize on europe-west2 for latency and operational consistency.
- Create Cloud Run services for **api** and **web**.
- Provision Cloud SQL PostgreSQL and enforce backup settings.
- Configure Secret Manager for broker, webhook, and alert credentials.
- Configure Artifact Registry and container naming conventions.
- Add password protection, role separation, and basic audit logging.
- Create Cloud Scheduler jobs for ingestion, retraining, and health pings.

Exit state

- Deployable cloud environment.
- Cost alerts live.
- Secrets stored correctly.
- Database reachable from the application layer.

Checkpoint 2 - Data model and storage conventions

Objective: define the data shape once so the platform does not become hard to trust later.

Tasks

- Create tables for bars, signals, regimes, alerts, macro features, event calendar entries, news articles, model runs, and backtests.
- Store raw source payloads alongside normalized records for auditability.
- Add idempotency keys for webhooks and scheduled jobs.
- Define a canonical timestamp policy in UTC.
- Define how historical bars, engineered features, and backtest outputs are partitioned and archived.
- Create a versioning rule for strategy definitions and model configurations.

Exit state

- Stable schema.
- Audit trail from source payload to dashboard output.
- Reproducible model and backtest storage path.

Checkpoint 3 - Market data and broker connector layer

Objective: establish the live and historical gold feed that everything else depends on.

Tasks

- Connect the chosen IBKR API mode and confirm authentication workflow.
- Pull live and historical bars for the chosen Gold instrument.
- Define broker session recovery and reconnect behavior.
- Add a fallback or redundancy path using MT5 or TradingView if required.
- Cache bars locally so repeated backtests do not hammer external APIs.
- Confirm all session windows: London open, New York open, London-New York overlap, and

LBMA Fix windows.

Exit state

- Reliable live bar ingestion.
- Usable historical store for indicators and research.
- Known session calendar and time handling.

Checkpoint 4 - Macro, event, and market context layer

Objective: connect the non-price drivers that make Gold different from generic chart trading.

Tasks

- Ingest FRED series for real-yield context and dollar context.
- Ingest BLS data and release metadata relevant to CPI, payrolls, and employment.
- Ingest BEA data relevant to GDP and PCE.
- Ingest CFTC positioning data for COT.
- Add an economic calendar feed for daily event risk.
- Add manual or CSV ingest for WGC reports and PBOC reserve updates.
- Add GLD holdings, VIX, oil, and GDX relative-performance feeds where feasible.
- Normalize all events and macro updates into a single context table the UI can query cleanly.

Exit state

- Fundamentals tab can be powered from structured data, not hand-entered text.
- The signal engine can tag headwind, tailwind, or mixed macro context.

Checkpoint 5 - News, sentiment, and NLP layer

Objective: add a low-cost qualitative layer without creating a second expensive research stack.

Tasks

- Add MarketAux for financial news, entities, and first-pass sentiment.
- Filter feeds to Gold, Treasuries, Fed, dollar, commodities, and geopolitics.
- Add a Gemini summarization pipeline for morning briefs, article clustering, and reason codes.
- Hard-cap token use and prompt counts so costs stay predictable.
- Add a human-review flag for major geopolitical items and policy headlines.
- Store article sentiment and model summary outputs separately from raw headlines.

Exit state

- Fundamentals tab can explain *why* the environment changed, not just that it changed.
- NLP remains a small variable-cost layer instead of a large subscription burden.

Checkpoint 6 - Signal engine MVP

Objective: get the first live, trustworthy signal path running.

Tasks

- Implement the existing EMA20 / EMA50 / RSI14 checklist rules for BUY, SELL, and NO TRADE.
- Add ADX, momentum direction, and regime classification so the signal stack is richer than the base checklist.

- Build confidence scoring from indicator agreement.
- Record every signal with timestamp, price, regime, and confidence.
- Add London Fix and session-state tags to each signal.
- Build no-trade filters around flat EMAs, RSI extremes, event proximity, and mixed macro conditions.

Exit state

- One stable, logged Gold signal engine.
- Every signal visible and reviewable in history.

Checkpoint 7 - Home tab delivery

Objective: make the system instantly useful before the deep tabs are complete.

Tasks

- Build the current regime card.
- Build the active signal card with confidence score.
- Add last regime switch and recent alert history.
- Add signal stack summary and system-health badges.
- Add last data-ingest, last model-run, and last broker-heartbeat timestamps.
- Add a compact watchlist for core correlated instruments.

Exit state

- One glance tells the trader whether to lean long, lean short, or stand aside.

Checkpoint 8 - Detailed Fundamentals tab

Objective: translate the trading manual's macro framework into a live research surface.

Tasks

- Add real-yield, DXY, Fed stance, CPI, NFP, GDP, and PCE panels.
- Add central-bank accumulation, PBOC update tracker, and WGC summary block.
- Add COT positioning and GLD flow context.
- Add VIX, Brent, and GDX relative-performance modules.
- Add a major-event board and same-day event risk flags.
- Add curated news blocks and short NLP summaries for Gold-relevant developments.

Exit state

- Fundamentals can be reviewed without leaving the platform.
- The macro backdrop, event risk, and central-bank context are no longer ad hoc.

Checkpoint 9 - Detailed Indicators tab

Objective: expose the chart-logic and execution filters that sit behind the signal.

Tasks

- Add M15, H4, and Daily structure summaries.
- Add EMA alignment, RSI, ADX, VWAP, session range, and distance-from-VWAP panels.
- Add spread, volume, open-interest, and stop-hunt context where data is available.
- Mark London Fix windows, New York handoff, and low-liquidity periods.

- Add the correlation-check panel: Gold vs real yields, DXY, GDX, VIX, and oil.
- Add the execution filter checklist that decides whether a setup is valid or should be ignored.

Exit state

- The trader can inspect the signal mechanics and not rely on a black box.

Checkpoint 10 - Projection, price modeling, and forecasts tab

Objective: add a forward-looking layer that is useful, interpretable, and cheap enough to keep.

Tasks

- Build baseline models first: naive, session expected move, rolling volatility bands, and at least one tree-based model.
- Create a feature set that combines market structure, macro deltas, event flags, and news sentiment.
- Schedule retraining with Cloud Run Jobs and store artifacts in Cloud Storage.
- Save forecast horizon, interval range, feature importance, and model confidence for every run.
- Add scenario narratives such as **bullish macro / weak chart**, **bearish macro / bullish divergence**, and **event-risk no-confidence state**.
- Add a clear rule for when forecasts are hidden because confidence is too low.

Exit state

- Forecasts become an explainable assistant layer, not an ungrounded prediction widget.

Checkpoint 11 - Backtesting tab

Objective: turn the manual's process into something measurable and repeatable.

Tasks

- Encode the base EMA/RSI checklist strategy.
- Encode Real Yield Divergence, London Fix Momentum, and New York Session Breakout strategies.
- Add slippage, spread, and event-risk assumptions.
- Add walk-forward testing and out-of-sample evaluation.
- Add parameter sweeps with sensible guardrails.
- Build outputs for hit rate, expectancy, average hold time, drawdown, regime split, and event-day split.
- Version every backtest run so later improvements can be compared honestly.

Exit state

- Strategy changes are judged by evidence, not memory.

Checkpoint 12 - Alerting, hardening, and launch readiness

Objective: make the platform reliable enough to trust on live trading sessions.

Tasks

- Add Telegram live alerts for new signals, regime switches, and high-confluence events.
- Add optional email alerts using existing mail infrastructure if required.
- Add auth hardening, rate limiting, and recovery runbooks.
- Test database backup and restore.
- Add cost dashboard and quota monitoring.

- Run a live observation period on Gold and record false positives, latency, and data gaps.
- Close launch only after the acceptance week has passed without critical failures.

Exit state

- Stable internal launch.
- Clear path to later commercialization if desired.

5 Master bullet-point task list

Product and UX

- ☐ Freeze version 1 scope.
- ☐ Freeze tab definitions and per-tab acceptance criteria.
- ☐ Define role-based access: admin, trader, viewer.
- ☐ Create a visual system for signal cards, macro cards, and model cards.
- ☐ Create empty states for missing data, stale data, and high-risk event windows.

Infrastructure

- ☐ Billing alerts and monthly budget caps.
- ☐ Cloud Run service templates.
- ☐ Cloud Run job templates.
- ☐ Cloud SQL creation, backups, and access rules.
- ☐ Secret rotation plan.
- ☐ Artifact retention and image cleanup.
- ☐ Logging and health dashboards.

Data and connectors

- ☐ IBKR connector and historical pull.
- ☐ Optional MT5 or TradingView redundancy.
- ☐ FRED ingest.
- ☐ BLS ingest.
- ☐ BEA ingest.
- ☐ CFTC ingest.
- ☐ Calendar feed ingest.
- ☐ MarketAux news ingest.
- ☐ WGC and PBOC manual ingest workflow.

Signal and research layer

- ☐ Implement EMA20 / EMA50 / RSI signal logic.
- ☐ Add ADX, momentum, and regime scoring.
- ☐ Add macro headwind / tailwind state.
- ☐ Add event-risk suppression logic.
- ☐ Add divergence detection against real yields and DXY.
- ☐ Add signal history and regime-switch history.

Forecasting and backtesting

- ☐ Create feature store.
- ☐ Create baseline forecast models.
- ☐ Add scheduled retraining.
- ☐ Add forecast logging and confidence control.
- ☐ Encode strategy library.
- ☐ Add walk-forward tests.
- ☐ Add slippage and session filters.
- ☐ Export reports and result snapshots.

Operations and launch

- ☐ Telegram alerting.
- ☐ Optional email routing.
- ☐ Restore test.
- ☐ Uptime and stale-data monitoring.
- ☐ Cost review at the end of the first live observation window.
- ☐ Post-launch backlog for non-Gold assets and commercialization ideas.

6 API and connector plan

Selected approach

Vendor minimization rule

The uploaded dashboard prototypes reference a separate metals API and FX API. Because an IBKR account already exists, the lower-cost plan is to **replace those paid market-price placeholders with IBKR market data where permissions allow**, and use public macro sources plus one low-cost news feed and one low-cost calendar feed. That keeps the stack lean and reduces duplication.

Source / API	Use now?	Monthly cost	Role in the platform
IBKR Web API / TWS API	Yes	\$0	Broker integration, account context, live and historical data where account permissions allow
IBKR market data add-on (only if needed for the chosen Gold instrument)	Maybe	\$0 to \$1.55+	Optional market-data subscription if the chosen Gold path requires it, especially futures
FRED API	Yes	\$0	Real yields and macro series ingestion
BLS Public Data API	Yes	\$0	CPI, labor, and payroll-related economic series
BEA Data API	Yes	\$0	GDP, PCE, and broader macroeconomic series
CFTC public reporting	Yes	\$0	COT positioning ingest
MarketAux Basic	Yes	\$29.00	Financial news, sentiment, entity tagging, and article metadata
EODHD calendar package	Yes	\$19.99	Economic event calendar and event metadata
Vertex AI Gemini 2.5 Flash Lite	Yes, capped	Target \$10 cap	Summaries, clustering, reason codes, morning notes, and forecast explanations
WGC reports / PBOC updates / issuer pages	Yes, manual ingest	\$0	Central-bank buying context, reserve updates, ETF holding context where manual ingest is acceptable
Telegram bot delivery	Yes	no new paid line item assumed	Immediate alert routing to the team
Optional EODHD EOD + Intraday package	Defer	\$29.99	Only add if IBKR historical coverage or rate limits prove too restrictive

What this means operationally

- News and sentiment do not need two separate paid vendors if MarketAux is used for first-pass analysis and Gemini is only used for summarization.
- Public macro APIs cover a large amount of the fundamentals tab for free.
- The biggest structural cost on Google Cloud is the database, not the stateless application layer.
- The biggest risk to the budget is uncontrolled scope creep, not baseline infrastructure.

7 Monthly cost budget

All figures below are **budget estimates**, not guaranteed invoices. They assume a small internal user base, same-region services, CPU-only modeling, modest storage, and no premium market-data expansion beyond

the chosen minimum. Exact spend will vary with traffic, logging, storage growth, broker permissions, and how often jobs run.

Recommended working budget

Google Cloud line item	Budgeted monthly amount	Planning note
Cloud Run web + API services	\$5.00	Low-traffic internal app, request-based billing, scale-to-zero target
Cloud Run Jobs for ingest / retrain / backtest refresh	\$2.00	Scheduled CPU jobs only
Cloud SQL PostgreSQL (db-g1-small) + 20 GB SSD + 20 GB backup	\$30.55	Recommended sturdier small database option
Secret Manager	\$0.00	Stay within free usage limits
Artifact Registry	\$0.10	Small private image footprint
Cloud Storage for artifacts / exports	\$0.40	Roughly 20 GB standard regional storage
Cloud Scheduler	\$0.20	More than three jobs assumed
Cloud Logging	\$0.00	Keep within free allotment
Google Cloud subtotal	\$38.25	

External API / data line item	Budgeted monthly amount	Planning note
IBKR API integration	\$0.00	Existing account assumed
IBKR market-data add-on for Gold (optional)	\$1.55	Only if the selected Gold instrument path needs it
MarketAux Basic	\$29.00	News + sentiment + entities
EODHD calendar feed	\$19.99	Economic events and release schedule
Vertex AI usage cap	\$10.00	Explicit hard cap for summaries and reason codes
External subtotal	\$60.54	

Total planning line	Amount	Comment
Selected stack total	\$98.79	Recommended working configuration
Contingency buffer	\$25.00	Leaves room for mild drift or temporary experimentation
Planned working budget	\$123.79	Comfortably below the \$200 ceiling
Headroom to ceiling	\$76.21	Available for growth before review is required

Lean alternative

If cost pressure becomes the main priority, the same plan can start on a **db-f1-micro** database. That takes the selected stack down to roughly \$80.90/month before contingency. The trade-off is that shared-core **db-f1-micro** and **db-g1-small** are not covered by the Cloud SQL SLA, so the lean path should be viewed as a startup convenience, not the final steady-state architecture.

What is intentionally not in the budget

- No GPU spend.
- No premium secondary market-data vendor.
- No BigQuery warehouse.
- No separate paid NLP platform.
- No separate paid email relay assumed in version 1.

8 Recommended final stack under the cap

Use this by default

- **Frontend:** Next.js on Cloud Run.
- **Backend:** FastAPI on Cloud Run.
- **Database:** Cloud SQL PostgreSQL, **db-g1-small** to start.
- **Jobs:** Cloud Run Jobs + Cloud Scheduler.
- **Broker / price feed:** IBKR first, MT5 or TradingView only as optional redundancy.
- **Macro data:** FRED, BLS, BEA, CFTC, plus manual WGC and PBOC ingest.
- **News / NLP:** MarketAux Basic + capped Gemini usage.
- **Events:** EODHD calendar feed.
- **Alerts:** Telegram first, optional email later.

Why this stack wins

- It fits the current GCP foundation.
- It removes unnecessary paid market-price APIs.
- It keeps the recurring bill well below the target.
- It still leaves enough room for modeling, news, and backtesting.

9 Open decisions that must be settled early

- Which Gold instrument is the source of truth for live trading and backtesting?
- Is TradingView kept as a visible signal-validation layer, or removed from the core path?
- Is the initial historical store broker-only, or do we add a secondary backup feed later?
- How much of the central-bank, ETF-flow, and Shanghai / China context is manual in version 1 versus automated later?
- Is email truly required in version 1, or is Telegram enough for the first live cycle?

10 Source alignment to the existing materials

This roadmap is not a greenfield invention. It is aligned to the uploaded materials in the following way:

- The product shape, alerting logic, and staged build pattern come from the dashboard brief.
- The current architecture recommendation follows the project summary that already places the system on Next.js, FastAPI, Cloud Run, and Cloud SQL.
- The initial signal logic starts from the existing EMA20 / EMA50 / RSI checklist.
- The fundamentals, correlations, session logic, strategy types, and daily review process come from the trading manual.
- The dashboard prototypes already reveal the intended fundamentals-card layout and the earlier placeholder data-source choices.

Appendix A - Pricing and source notes

Exact URLs are kept in the LaTeX source file so the printed appendix stays readable.

Code	Source title	What it was used for
G1	Google Cloud Run pricing	Request-based free tier, per-vCPU and per-GiB-second pricing, and same-region service assumptions
G2	Google Cloud SQL pricing	Shared-core instance pricing, storage pricing, backup pricing, network notes, and SLA caveat
G3	Secret Manager pricing	Active secret version pricing, access-operation pricing, and free-tier limits
G4	Artifact Registry pricing	Private container-image storage pricing and free allowance
G5	Cloud Scheduler pricing	Per-job pricing and free-job allowance
G6	Cloud Storage pricing	Regional storage pricing in London for artifacts and exports
G7	Google Cloud Observability pricing	Logging free allotment and storage-pricing guardrails
G8	Vertex AI Generative AI pricing	Gemini 2.5 Flash Lite usage pricing and grounded-prompt warning
I1	IBKR Web API and market-data docs	Free API availability and market-data subscription requirements by instrument type
I2	IBKR market-data pricing	Optional COMEX real-time pricing reference
I3	IBKR Gold product pages	Confirmation that IBKR supports spot-gold products relevant to planning

Code	Source title	What it was used for
N1	MarketAux pricing and product page	News feed cost, daily request allowance, and sentiment/entity capability
N2	EODHD pricing	Calendar feed cost and optional intraday backup-feed pricing
P1	FRED API docs	Public macro API reference for real-yield and economic-series planning
P2	BLS Public Data API docs	Public labor and inflation API reference
P3	BEA Data API	GDP and PCE API reference
P4	CFTC public reporting	Public COT reporting reference

Appendix B - Source alignment to uploaded files

Uploaded source	Used for
DashboardProjectBrief.pdf	Product goals, alert requirements, staged build pattern, and original dashboard concept
PRINCIPIA_PROJECT_SUMMARY.md	Current technical baseline on Google Cloud, already-provisioned architecture, and existing deployment direction
Principia_Gold_Signal_Checklist.pdf	Initial M15 EMA / RSI signal rules and no-trade conditions
Principia_XAUUSD_Trading_Manual.pdf	Gold macro drivers, strategy library, intraday session logic, and daily checklist framework
gold_fundamentals_dashboard*.html	Fundamentals-tab card design, card hierarchy, and early placeholder data-source choices